# Machine Exercise 3

# LSTM and NARX System Identification

*In this machine problem you will model a linear system using an LSTM network. Each student is assigned a dataset and a model of a linear system that generated the dataset. The performance of the LSTM models is compared with the Neural State-Space model. The nonlinear Newell-Lee evaporator system is also modeled using an NARX structure.*

Machine Exercise

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## 3.1 Instructions

**INSTRUCTIONS**. This is an open-notes, open-books exercise to be done individually. You are to submit a lab report detailing the process and analysis of your chosen approach, and an executable Jupyter notebook or a MATLAB Live script `.mlx` file. Make sure to write your name and student number in the first part of your submission so that it is easily identifiable to you. Include figure, plots, diagrams, tables, etc. as necessary in your discussion to clarify your answers and analysis of the results.

From your `.mlx` or `.ipynb`, generate the PDF output with comments, plots, answers to the questions and discussion. Submit as a single zipped file the `.mlx` or `.ipynb` and the the PDF output. If there are other external files needed (*.csv, *.wav, etc) include everything in the zipped file and upload to the UVLe submission bin before the deadline. Anything submitted after the deadline can still be graded but with a 0.6 multiplier penalty to the final score. The submitted zipped file must be easily identifiable to you, use the suggested file name `<Surname>_AI321_ME03.zip` when submitting any machine exercise.

## 3.2 The dataset is different for each student

a. You will generate two types of APRBS signals (we will designate as small and large) that will be used as inputs to the system: (1) small amplitude and variance and (2) large amplitude and variance. The number of samples is 20,000.

b. Using the two (2) APRBS signals as input to the linear system, generate two (2) sets of output datasets. Foe each input and output pair, use the first 10,000 samples (50%) as the training set and the samples from 10,001 to 20,000 as the validation set.

c. Generate two LSTM model structures: (1) to model the system with small APRBS input and (2) to model the system with large APRBS input. Estimate the parameters of the two LSTM models using the training set and validate using the validation set using the metrics: MSE, NMSE and $R^2$.

d. Compare the performance of the two LSTM models. Which one is better and why?

e. Compare the performance of the LSTM model with a Neural State-Space Model. This model structure is supported in the System Identification Application and Toolbox.

f. There are twelve linear systems assigned below, one per student (`s1.mat` through `s12.mat`).

| Student | Model Name | Student | Model Name |
|---|---|---|---|
| Asares | sys1.mat | Benzon | sys2.mat |
| Bolo | sys3.mat | Dayo | sys4.mat |
| De Guia | sys5.mat | Francisco | sys6.mat |
| Ignacio | sys7.mat | Mendoza | sys8.mat |
| Munar | sys9.mat | Renovalles | sys10.mat |
| Rosell | sys11.mat | Carcillar | sys12.mat |

**Table 3.1:** Student-to-dataset assignment for MP3 (12 students).

g. **BONUS.** You can use any of the deep learning models to estimate the parameters of the unknown state-space system, if you think the previous SIM and PEM models are insufficient for the task. Why did you choose a particular Neural network / Deep learning

model and report on the performance of the your chosen model against the SIM, PEM and LSTM methods.

## 3.3 Using LSTM to Model an Unknown Linear Model

In the previous exercise you obtained the best model using Subspace Identification and Predictive Error Models of this Single-Input Single-Output (SISO) system. In this machine problem you will try using LSTM networks and a Neural State-Space Model.

- **LSTM Networks and Neural Sate-Space Models**. Use an LSTM network as the model structure to determine the parameters of the linear dynamic system. You are free to specify the parameters of your LSTM model as you see fit but be aware of our philosophy of parsimony in system modeling, we favor the system with least number of parameters. The simpler the model the better. In this exercise you will use two different sets if APRBS signal to generate your own training and validation datasets. Train and validate the performance of two LSTM model using two different sets of APRBS signals and report its performance. Compare the performance of the two LSTM models with your previous models using subspace identification and predictive error models.

If you think that you need more than 10,000 samples of data to train the LSTM network, you can generate more samples, since you are given the system model that generated the datasets. Report if you need to generate training and validation datsets that is more than 20,000 samples. Explain why 20,000 samples is insufficient for your chosen model structure.

Loading the .mat file into the MATLAB workspace will load the model which you can use to generate more input-output pairs. You can instantiate a different set of APRBS signal and increase the duration of the data to create a different set of training and validation data using the model. One way to train a model is to create an APRBS data set with small input variations, meaning small variance and amplitude, and create another APRBS data set with large amplitudes and larger variability, meaning large variance and amplitude. Create a separate training and validation set and test the LSTM model prediction performance on the simulated APRBS data. Generate both clean output data and a noisy output data corrupted by noise with a fixed variance, using a default variance of $\sigma^2 = 0.01$.

### Performance Analysis

- Test the $R^2$ measure of the LSTM with the clean output data, and with the noisy output data.
- Determine the $R^2$ performance of the model in making a one-step prediction of the output from the past values of the data.
- Plot the autocorrelation of the output residual, the prediction error, and the cross-correlation of the input and output residuals. What information does the autocorrelation, and the cross-correlation have about the performance of the model prediction?
- Determine the $R^2$ performance of the model in making a five-step prediction of the output from the past values of the data.

For a tutorial on how to use LSTM model for model identification of linear systems, one is available in our UVLe page.

- **Neural State Space Models**. A Multi-Layer Perceptron can also be used to model the linear system dynamics. Instead of an LSTM, use an MLP to determine the parameters of the linear dynamic system assigned to you. Like the LSTM exercise above, you are free to generate a different set of training and validation data, if you think the given input-output pairs are insufficient.

- Generate a clean output and a noisy output, assess the performance in terms of residual error, $R^2$, one-step and 5-step prediction accuracy of the neural state space model on the clean output, and also on the noisy output.

- **LSTM vs Neural State-Space Model**. Make a performance analysis comparison of the LSTM and the Neural State-Space output in terms of model fit, one-step ahead prediction, five-step ahead prediction, residual signal autocorrelation and cross-correlation, number of model parameters, etc.

## 3.4 Newell-Lee Evaporator Model

Using the Newell-Lee evaporator model, this system exhibits linear and non-linear behavior. The model is simulated, and the input output pairs are in the associated data file "p2_dat.mat". This new data file is the result of simulating the system with a longer simulation time.

Split the data into training and validation. The first 0 to 5000 data points is the training set, the 5001 to 6001 data points is the validation set.

Estimate the system model starting with NARX model structure. Estimate the model parameters using the train data and test the performance on the validation data. In model estimation determine the following:

- The best model order according to the metric for small amplitude inputs on the training data, just to check how well the model performs, then check the model performance on the validation data.

- The best model order according to the metric for large amplitude inputs on the training data, just to check how well the model performs, then check the model performance on the validation data. Show the plots of the output and model prediction on the training data and the validation data.

- Change the model estimation algorithm to Hammerstein-Weiner Model and investigate the different non-linear model structures which will give a good model prediction. Compare the performance NARX model and the Hammerstein-Weiner Model structure that you have chosen.

- Use a Neural State Space Model as the model structure and do a performance analysis compared with the Hammerstein-Weiner Model.

- Use a State Space ARX model structure and do a performance analysis of the different models, the Neural State-Space Model and the Hammerstein-Weiner Model.

- Use an LSTM network model structure and compare its performance. Compare the performance of the non-linear models used: Non-linear ARX, Hammerstein-Weiner model, Neural State-Space model, State-Space ARX model and LSTM.

- What insights can you draw having seen the performance of these model structures?

## Reflection and Academic Integrity Declaration

This section is required and graded on the depth and specificity of your answers, not on whether or how much you used AI. This responses in this section can be considered when grading the exercise as a whole.

**AI 321's policy on AI tools — transparency, not prohibition**

You are free to use AI tools (Claude, ChatGPT, Copilot, etc) to help you learn, debug, and understand concepts in this course. What you can not do is submit AI-generated code, derivations, or written answers as your own work without disclosing them and demonstrating you understand them. Doing so is academic dishonesty, equivalent to plagiarism, whether or not the code runs correctly.

### 1. Approach Summary

In 3-5 sentences, describe what you did and why, in your own words, as if explaining it to a classmate who has not seen this exercise.

[*Response*]

### 2. Key Insight

What is one result or observation from this exercise that surprised you, or didn't match your initial intuition?

[*student response*]

### 3. Most Difficult Part

What part of this exercise took the longest to get right, and how did you resolve it?

[*student response*]

### 4. Disclosure on the use of AI systems

List every part of this exercise where you used an AI tool, one row per use. Write "None" in the first row if you did not use AI at all. The last column is the part that can be considered in grading this exercise. Be clear on your statements, on how AI actually helped you if it did. They do make mistakes, you can also list here the instances where the AI failed and you were able to recognize its failure.

| **Section / task** | **Tool used** | **What you asked it for** | **Explain the result in your own words, as if teaching it** |
|---|---|---|---|
| *e.g., Problem 2, debugging* | *e.g., ChatGPT* | *e.g., fix a shape mismatch in my autocorrelation matrix* | [*student response*] |

### 5. Self-Assessed Understanding

For each objective or major problem in this exercise, rate your confidence (Confident / Mostly confident / Still shaky) and add one sentence why.

| Objective / problem | Confidence | Why (one sentence) |
|---|---|---|
| | | |

## 6. Academic Honor Pledge

By submitting this exercise, I hereby affirm upon my honor the following:

1. The analysis, results, codes, answers to questions, discussions and conclusions in this machine exercise reflect my own understanding and work.
2. I used my own official student number whenever it is required, and the results generated were computed from my own personalized parameters/data, and not from anyone else.
3. I have disclosed fully and accurately any use of AI tools, LLMs, AI agents or similar systems in Section 4 above, and I can explain, in my own words, every part of what I submitted if asked.
4. I understand that submitting AI-generated work as my own can be considered as plagiarism and is an academic dishonesty, and that I abide with the University's policy on Academic Integrity as written in the Student's Code of Conduct.

**Signed (name):** _____ **Student number:** _____

**Date:** _____